

Predicting Future Aspects of Hospital Readmissions using a Sequence-Based Markovian Approach

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Capstone Project Report
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Abstract—The problem of hospital readmissions is very severe in the United States: currently one in five patients is readmitted to the hospital within 30 days of being discharged. The high number of hospital readmissions is indicative of gaps in quality of patient care. In addition, they place a huge financial burden on the healthcare system. Measures that can reduce the number of unnecessary hospital readmissions are therefore very valuable.

In this project we build machine learning models that can automatically flag patients who, at the time of hospital discharge, are likely to be readmitted within the next 30 days. We also predict their length of stay (LOS) and cost of the next readmission. Care providers can then carefully consider which interventions would be appropriate to reduce the risk of readmission of the flagged patients. We train our machine learning models on historical patient data of the California Office of Statewide Health Planning and Development (OSHPD)¹. Specifically, we apply a sequence-based approach using N-grams, Hidden Markov Models and Markov Chain Models to predict 30day risk of readmission, the length of stay and cost of the next readmission of a patient. These sequence-based approaches take into account all the previous admission records of a patient to predict whether that patient will be readmitted within 30 days of most recent discharge and what the length of stay (LOS) and cost of that readmission will be. We evaluate these sequence-based prediction models by comparing their accuracy to that of a majority baseline prediction algorithm.

I. INTRODUCTION

There are about 34 million hospital admissions annually in the United States [2]. Out of these, there are many cases where a patient gets readmitted to the hospital after being discharged. This problem is so

severe that currently one in five patients is readmitted to the hospital within 30 days of being discharged [9]. Surprisingly, many of these readmissions are actually avoidable. This problem is important both from the point of view of patient care and finance [10]. If we can predict the possibility of readmission within 30 days², many of the readmissions can be prevented if appropriate interventions are designed and administered in time [8]. In addition to exacerbating access to care issues, hospital readmissions are indicative of gaps in care coordination.

Our aim is to design and develop a robust predictive model for readmission reduction. Given state of the patient at discharge, using sequence based machine learning, our goal is to: (a) predict the likelihood of a 30day readmission, (b) the corresponding length of stay (LOS) during that next episode, and (c) the cost of such a readmission. If we predict that indeed the patient's next readmission will be within 30 days of the last discharge, then that patient is considered to be a high risk patient and healthcare providers need to take steps to avoid such a readmission.

Our Machine Learning experiments followed a twofold strategy: First we develop a readmission prediction model without the assumption that we know what type of chronic patient a person might be; termed as the 'AllCause' model [12]. Then we focus only on CHF population, given the complexity of CHF - a frequently observed cause of hospital admissions globally [3].

²We use 30 days as a threshold to predict hospital readmission, because it has been identified as a performance measure for institutions and a target of reimbursement-based improvement incentives that will come to bear in the near future [5]. In the context of Medicare, readmissions have been generally defined as a patient being hospitalized within the next 30 days of current hospital stay [4].

¹<http://www.oshpd.ca.gov/>

In this project, we use the patients' hospital admission data provided by the California OSHPD (Office of Statewide Health Planning and Development). This dataset consists of records for patients discharged from California-licensed hospitals over five years (2009-2013). Each admission of a patient is a separate record in the dataset. A patient record includes demographic information about the patient, such as patientID, age, gender, birthday etc. and also admission information like primary diagnosis, secondary diagnoses, admit and discharge dates among others. We use selected information from this patient information to predict whether the patient will be readmitted to the hospital within 30 days and what the LOS and cost of that readmission will be.

It is possible to solve our three prediction tasks (30day, LOS and cost) using event-based prediction and sequence-based prediction [7] [12] [16] [17]. The main distinction between the two is: in event-level prediction, only the details of the current hospital admission are considered when making the prediction of readmission, while in sequence-based prediction, we consider the full admission history of the patient to predict the 30day readmission of the patient, LOS and cost of the readmission. Existing work on 30day readmission prediction is primarily done at event-level because that is conceptually simpler. Taking the full admission history of patients into account is more challenging from a machine learning perspective but can potentially lead to more accurate predictions. In this project, our primary focus will be on the development of a sequence-based approach to solve the three prediction tasks (30day, LOS and cost), i.e we will link all the admission records of one patient together and predict whether the next admission will be within 30 days, the LOS and cost of the readmission, using the information obtained from the sequence. An example of how we build sequences is shown in Section II-C.

We will mainly build our models using three techniques: N-gram [14], Hidden Markov Model and Markov Chain Model [11]. To predict the next value for a given test sequence using N-grams, probabilities are estimated during the training phase, which are then used during the testing phase to predict the next most likely symbol.

A Hidden Markov Model (HMM) [15] is a statistical Markov Model in which the system being modeled is assumed to be a Markov Process with unobserved (hidden) states. In simpler Markov Models (Markov Chains), the state is directly visible to the observer, and therefore the state transition probabilities are the only parameters. In a Hidden Markov Model, the state is not directly visible, but the output, dependent on the state, is visible. In this study we obtained results using Multinomial HMMs, i.e. HMMs with discrete emissions [1]. Internally, the probability of each symbol when appended at the end of each sequence is obtained and

the symbol with maximum probability is predicted.

A Markov Chain is a first order Markov model. Such models are considered stateless or memoryless because they assume that future states depend only on the current states and not on the states that preceded it. A Markov chain has a set of states, $S = \{s_1, s_2, \dots, s_r\}$. The process starts in one of these states and moves from one state to another. The probability of movement from state s_i to s_j is p_{ij} . p_{ij} is called the transition probability. The process is characterized by a state space, a transition matrix describing the probabilities of particular transitions, and an initial distribution across the state space.

To outline the flow of this report, we will first describe the dataset used in this project in Section II. Then we will explore the above mentioned techniques without using any information from the discharge records in section III. We will only use the corresponding label of the problem we are trying to predict. Eg: if we want to predict the LOS of the next readmission, we will only use the LOS values in the discharge records for all previous admissions. Later, we will incorporate information from the discharge records, like severity code of the admission, LOS, age and gender of the patient in section IV. Lastly, we will compare all our models and decide which ones work best for each prediction task.

Part of this work has been presented at ACM-BCB 2016 [6]. It contains preliminary progress on the Ngram and Multinomial HMM models. The corresponding paper and poster are appended to this report.

II. DATA ENGINEERING AND FEATURE CONSTRUCTION

In this section, we will describe the original OSHPD data, and the method we used to transform that data into sequences without features (OSHPD-SPICE³ data).

A. OSHPD data

In this project, we apply machine learning techniques to predict 30day risk of readmission (30day), length of stay (LOS) and cost (COST) of admission on admission data of patients provided by the California OSHPD (Office of Statewide Health Planning and Development). This dataset consists of records for inpatients discharged from California-licensed hospitals over five years (2009-2013).

The data contains about 15 million rows, and each admission of a patient is a separate record in the dataset. The data has about 200 fields, which include personal information about the patient, like PatientID, Age, Gender, birthday etc. and also admission information like primary

³SPICE in this context refers to the format of data used in the Sequence Prediction Challenge (SPICE) competition 2016. <http://spice.lif.univ-mrs.fr/index.php>

diagnosis, secondary diagnoses, admit and discharge dates, length of stay, and cost of stay among others. The fields primary diagnosis and secondary diagnoses contain information in the form of ICD9 codes, which are six-character alphanumeric medical codes. The data in this raw form contains a lot of inconsistencies and hence we preprocess it. The preprocessing steps are described in subsection B [13].

We extracted all the Congestive Heart Failure Patients (CHF) in the data into a separate file. This is because we wanted results of only CHF patients separately, since CHF is a very common reason for patient readmission. We extracted 20% of patients for testing and kept 80% for training. Thus, after all the preprocessing steps, we now have four data files: ALLCAUSE-PUBLIC (this the training file for the data which contains both CHF patients and patients for all other disease categories), CHF-PUBLIC (this dataset is the training file which contains CHF patients only), ALLCAUSE-PRIVATE (testing file for ALLCAUSE dataset), CHF-PRIVATE (testing file for CHF dataset).

We converted the LOS and COST field values in the whole data into buckets (Shown in Tables I and II) in order to convert a regression task into a classification task. We do this because our models in this project either do not perform well for regression tasks or run purely on classification tasks. LOS buckets were chosen as defined in the standardized readmission tool LACE. COST buckets were created by using the first quartile, median and third quartile as boundaries. Since these boundaries are different for ALLCAUSE and CHF data, we have different ranges for COST buckets for these two groups. After bucketing, we have 6 buckets for LOS (1-6) and 4 buckets for COST (1-4). The 30day field (represents whether the patient will be readmitted within 30 days or not) already had binary values (0,1) so we did not need to convert that field into buckets.

B. Preprocessing

We preprocess the raw OSHPD data in the following way:

- Merging all years data: The raw data from OSHPD is divided into 5 years from 2009 to 2013. The first step is to merge the data from different years, so that we have the whole data in a single database. After merging, the data has about 15 million rows and about 200 columns.
- Sorting the data: The merged OSHPD data will then be exported into a csv file and ordered by rln and admt_date. This will group the data of each patient together by admit date.
- Cleaning inconsistencies in gender and race_grp: This demographic information should remain the

same for all the readmission records of the same patient. We need to remove mismatches, by assigning them the majority value. Also, the value '2' (Unknown) in gender will be replaced by '0' (Male) which is the majority value in the dataset.

- Merge transfer admissions: There are scenarios where the next admission date is earlier than the current discharge date or the next admission date is on the same day of the current discharge date. We need to merge these records carefully without losing information.
- Construction of response variable: There is no column to denote the class label for our problem. We need to create a column 'thirtyday' to store whether that patient was readmitted within 30 days of being discharged from the current admission. This column will hold a boolean value, 1 or 0. We will assign a value 1 if the admission date of the next admission of the same patient is within 30 days of the current discharge date, 0 otherwise.
- Grouping diagnosis and procedure codes into CCS: CCS codes collapse the ICD9 codes into broader categories. We make additional columns for CCS codes and assign a 1 if the patient has an ICD9 diagnosis or procedure code that falls under that CCS code. Specifically, we will remove the 500 odiag and oproc columns, and a series of new columns with names starting with DXCCS and PXCCS will be created to denote each diagnosis and procedure code in CCS. For each admission, we read through all ICD9 diagnosis and procedure codes, and set 1 for each DXCCS and PXCCS columns with corresponding CCS codes.
- Remove columns with zero variance: There are some columns which contain only empty or constant values and will not give us any information to solve our problem. We will remove such columns.
- Remove records of patients who have not been admitted more than once: When patients have only one hospital encounter, we will remove those records since there is not enough information to form a sequence and predict their next readmission.

C. OSHPD-SPICE data

We transform the original OSHPD data with all the features, into a form which contains only sequences of symbols of the problem we want to predict. This format of data contains header of a form such as: '13546 2'. This means that the data contains '13546' sequences/patients and the individual sequences contain 2 different symbols. Each sequence is prepended by the length of that particular sequence.

To elaborate on how the new data was formed, we take, for example, the admission history of a patient

TABLE I
LENGTH OF STAY (LOS)-VALUES TO BUCKETS.

LOS Buckets	LOS Values (Days)
1	1
2	2
3	3
4	4-6
5	7-13
6	14+

TABLE II
COST-VALUES TO BUCKETS.

Cost Buckets	Cost Values (ALLCAUSE)(\$)	Cost Values (CHF)(\$)
1	0-19,080	0-27,660
2	19,080-35,690	27,660-49,810
3	35,690-70,450	49,810-98,500
4	70,450+	98,500+

Anna (Table III). Given her admission history, while predicting risk of readmission (30day), we convert her admission records to a sequence of the form: ‘5 0 0 1 0 1’. This symbolizes that the patient has 5 hospital admissions (1st symbol in the sequence), 2 of which were within 30 days (symbol ‘1’). Similarly, we create sequences for LOS, namely ‘5 4 2 4 1 6’, and for cost, ‘5 4 3 3 1 4’. The OSHPD data has actual LOS (Days) and COST (\$\$) values as shown in Anna’s admission history. We convert those values to symbols in the sequence using Tables I and II.

We form 6 training and 6 test datasets of this form:

CHF_LOS_PUBLIC_SPICE.txt
 CHF_LOS_PRIVATE_SPICE.txt
 ALLCAUSE_LOS_PUBLIC_SPICE.txt
 ALLCAUSE_LOS_PRIVATE_SPICE.txt
 CHF_30DAY_PUBLIC_SPICE.txt
 CHF_30DAY_PRIVATE_SPICE.txt
 ALLCAUSE_30DAY_PUBLIC_SPICE.txt
 ALLCAUSE_30DAY_PRIVATE_SPICE.txt
 CHF_COST_PUBLIC_SPICE.txt
 CHF_COST_PRIVATE_SPICE.txt
 ALLCAUSE_COST_PUBLIC_SPICE.txt
 ALLCAUSE_COST_PRIVATE_SPICE.txt

For instance, CHF_LOS_PUBLIC_SPICE.txt would have training sequences of Length of Stay symbols from the original OSHPD CHF dataset, ALLCAUSE_COST_PRIVATE_SPICE.txt would have test sequences of COST symbols from the original OSHPD ALLCAUSE dataset.

We thus obtain a new form of OSHPD data, in which we only consider sequences of length four or more

(patients who have four or more admission records) since we need a minimum number of admission records for each patient to build a reliable model. The data statistics for this data are shown in table IV.

III. TECHNIQUES AND RESULTS - EXCLUDING FEATURES FROM DISCHARGE RECORDS

In this section, we describe the methods run on the OSHPD-SPICE data without using any features from the discharge dataset, and we see which methods perform better than the others.

A. Majority Baseline

The majority baseline method first obtains the majority label from the training data. This is the symbol which occurs most frequently at the end of sequences⁴. We naively predict this majority label for all test instances, then compare it to the actual test instance symbols and obtain the accuracy of this model. This model is the baseline, because this is a very naive way to predict, and our intelligent models should ideally perform better than this model.

B. N-gram

In the N-gram approach, we add the symbol -1 in front of all the training and test sequences to mark the beginning of a sequence. To predict the next value for a given test sequence using N-grams, probabilities are estimated during the training phase, which are then used during the testing phase to predict the next most likely symbol.

⁴The majority labels for the three problems (for both ALLCAUSE and CHF) are: 30day - 0, LOS - 4, COST - 4

TABLE III
ADMISSION HISTORY OF AN EXAMPLE PATIENT ANNA.

Admission Date	30day	LOS	Cost	Severity Code
Jan 22,2009	0	6	\$75099	2
Mar 30,2012	0	2	\$37531	0
Apr 15,2012	1	5	\$49657	0
Oct 1,2013	0	1	\$14107	1
Oct 20,2013	1	14	\$259616	2
??	??	??	??	??

TABLE IV
DATA STATISTICS

Problem	Num of Seq	Min-Max Seq length	Avg Seq Length (Std Deviation)
ALL-TRAIN	390,900	4-255	7.0 (5.3)
ALL-TEST	43,475	4-155	7.0 (5.1)
CHF-TRAIN	168,576	4-255	7.5 (5.6)
CHF-TEST	18,534	4-164	7.5 (5.5)

Training Phase: The frequencies of all possible grams (1-gram, 2-gram, etc.) in the training sequences are computed and stored in a dictionary. E.g. the training sequence “-1, 2, 1, 0, 3, 2, 0” will lead to the following frequency increments (in the dictionary for the 3-gram model): (-1):1, (2):2, (1):1, (0):2, (3):1, (-1,2):1, (2,1):1, (1,0):1, (0,3):1, (3,2):1, (2,0):1, (-1,2,1): 1, (2,1,0):1, (1,0,3):1, (0,3,2):1, (3,2,0):1

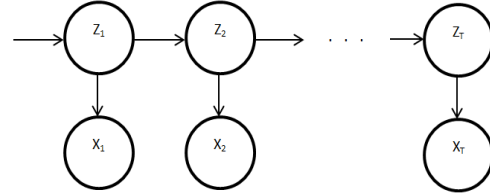
Testing Phase: Let L denote the length of the test sequence. In the N -gram approach, we take the last M symbols of the test sequence, with $M = \min(L, N-1)$, and we complete this to the most probable $M+1$ gram. i.e., given the last M symbols $(a_1, \dots, a_M)$, we look up the most frequent $M+1$ gram (N -gram) $(a_1, \dots, a_M, b)$ and predict b as the next symbol in the sequence. E.g., the 5-gram approach will consult all 5-grams from the dictionary whose prefix matches the last 4 symbols of the test sequence. It will only resort to grams of a smaller length if the test sequence has less than 4 symbols, and it will never consult any grams of length > 5 .

Table V shows results for different N -grams (2-gram to 6-gram) on OSHPD-SPICE data. From this table, we can conclude that increasing longitudinal information (longer sequence, hence larger N) will not necessarily improve the performance. The performance of the 4-gram models is comparable to, and sometimes even better than, that of 5 and 6-gram models.

C. Multinomial Hidden Markov Model

A Hidden Markov Model (HMM) [15] is a statistical Markov Model in which the system being modeled is assumed to be a Markov Process with unobserved (hidden) states. In simpler Markov Models (Markov

Fig. 1. Structure of a simple Hidden Markov Model



Chains), the state is directly visible to the observer, and therefore the state transition probabilities are the only parameters. In a Hidden Markov Model [1], the state is not directly visible, but the output, dependent on the state, is visible. Fig 1 below shows the structure of a very simple Hidden Markov Model, where X_i are the observed states and Z_i are the hidden states.

In this study we obtained results using Multinomial HMMs, i.e. HMMs with discrete emissions. We used the ‘MultinomialHMM’ library from the python ‘hmmlearn’ module. This library implements HMM with discrete emissions [1]. Internally, the library implements the Expectation Maximization (EM) algorithm and obtains the probability of each symbol when appended at the end of each sequence. The symbol with maximum probability is predicted.

Table VI shows the detailed Multinomial HMM runs recorded over different numbers of states and iterations. We take 5 runs of each and then average the results of all the 5 runs to obtain the final accuracy. We observed that increasing the number of hidden states and iterations was computationally very intensive and time-consuming to train the models. In addition, there was not a very large

TABLE V
N-GRAM RESULTS (ACCURACY)

Model	ALL-30	CHF-30	ALL-LOS	CHF-LOS	ALL-COST	CHF-COST
Majority Baseline	0.648	0.623	0.255	0.258	0.356	0.325
2-gram	0.621	0.606	0.266	0.258	0.381	0.336
3-gram	0.659	0.640	0.272	0.267	0.408	0.342
4-gram	0.659	0.640	0.273	0.269	0.418	0.348
5-gram	0.662	0.642	0.274	0.269	0.416	0.353
6-gram	0.662	0.642	0.271	0.262	0.417	0.347

Fig. 2. Accuracy of Multinomial HMM on varying % of training data for 30day risk of readmission problem.

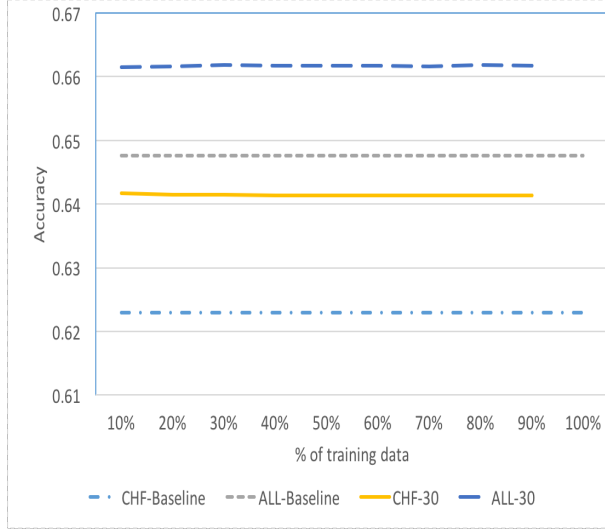
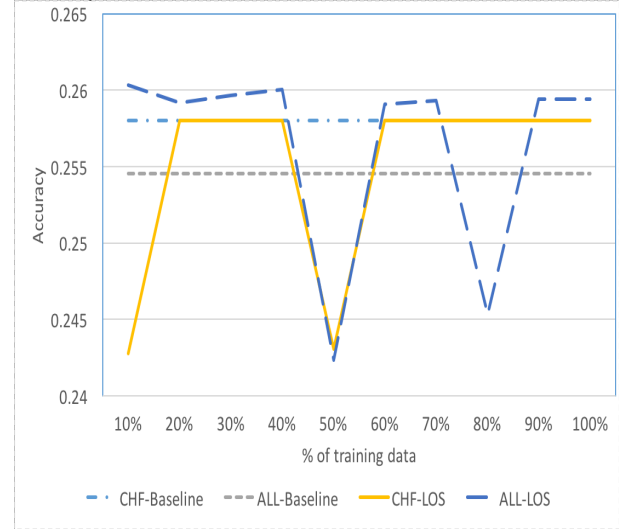


Fig. 3. Accuracy of Multinomial HMM on varying % of training data on LOS problem.



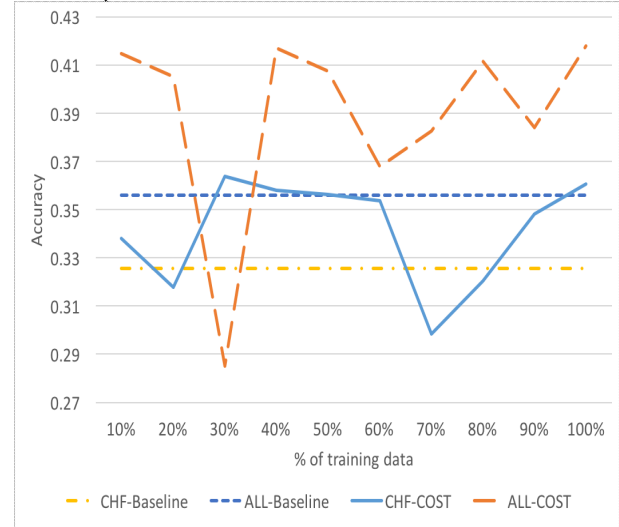
difference in the accuracy numbers when the number of hidden states and iterations were varied. Hence, for the HMM experiments in the remainder of the report, we fix the number of hidden states = 2 and number of iterations = 50 for all six problems as shown in Table VII

Fig. 2 - 4 show the accuracy of Multinomial HMMs trained on a varying percentage of the OSHPD-SPICE data. We calculated the accuracy on different percentages of training data with the objective of training only on a part of the full training data, since training on the full data was computationally intensive and time-consuming. We observed that there was a lot of fluctuation in accuracy numbers and we could not rely on a part of the training data to give us accurate results. Thus, we eventually train on 100% training data in this project.

D. Markov Chain Model

In this project we built a Markov Chain model⁵ that can leverage sequence information for predicting 30day risk of readmission, length of stay and cost of

Fig. 4. Accuracy of Multinomial HMM on varying % of training data on COST problem.



⁵<http://people.math.aau.dk/~jm/courses/PhD06StocSim/Chapters2.4.5.pdf>

TABLE VI
MULTINOMIAL HMM RESULTS FOR DIFFERENT NUMBERS OF HIDDEN STATES AND ITERATIONS. RESULTS ARE PRESENTED IN TERMS OF ACCURACY AVERAGED OVER 5 RUNS.

Model	ALL-30	CHF-30	ALL-LOS	CHF-LOS	ALL-COST	CHF-COST
Majority Baseline	0.648	0.623	0.255	0.258	0.356	0.325
MultinomialHMM States = 2 Iterations = 50	0.652	0.634	0.222	0.238	0.405	0.341
MultinomialHMM States = 4 Iterations = 50	0.588	0.613	0.209	0.227	0.378	0.325
MultinomialHMM States = 6 Iterations = 50	0.592	0.577	0.214	0.217	0.392	0.317

TABLE VII
FINAL ACCURACY, NUMBER OF HIDDEN STATES AND ITERATIONS FOR MULTINOMIAL HMM ON ALL PROBLEMS.

Model	ALL-30	CHF-30	ALL-LOS	CHF-LOS	ALL-COST	CHF-COST
Majority Baseline	0.648	0.623	0.255	0.258	0.356	0.325
MultinomialHMM Accuracy	0.652	0.634	0.222	0.238	0.405	0.341
Number of states	2	2	2	2	2	2
Number of Iterations	50	50	50	50	50	50

readmission. We use the OSHPD-SPICE data for this purpose, which contains only sequences of symbols of the problem we want to predict. We will use a first order Markov model. Such models are considered stateless or memoryless because they assume that future states depend only on the current state and not on the states that preceded it. We consider a first order Markov model because it enables computation that would otherwise be intractable⁶.

A Markov Chain has a set of states, $S = \{s_1, s_2, \dots, s_r\}$. The process starts in one of these states and moves from one state to another. The probability of movement from state s_i to s_j is p_{ij} . p_{ij} is called the transition probability. The process is characterized by a state space, a transition matrix describing the probabilities of particular transitions, and an initial distribution across the state space.

In this model, we form the state transition matrix with number of rows and columns equal to the number of symbols N in the problem we need to predict. There will be N such matrices corresponding to each of the symbols.

Markov Chain Model Example: We now consider an example to show how a Markov chain model could be applied to solve the 30day risk of readmission problem.

We know that our problem is a classification task with possible values (0,1). Thus we will construct two state

transition matrices: one for class label 1, where we will store the transition frequencies of training examples of all the patients who have class label 1 at the end of the sequence, and another matrix for class label 0, where we will store the transition frequencies of training examples of all the patients who have class label 0 at the end of the sequence. Eg:, for an example sequence ‘0 1 1 1 0’, we will store all the frequencies in the matrix for class label 0, since that is the last symbol in the sequence. The transitions ‘0-1’ (row 0 and column 1) and ‘1-0’ (row 1 and column 0) will have frequency 1, whereas, the transition ‘1-1’ (row 1 and column 1) will have frequency 2. We then perform Laplace smoothing on all the matrices.

When we want to predict a new instance, we append each possible symbol (0,1 for 30day risk prediction) at the end of a test sequence and obtain the probability of the last transition from the matrix which corresponds to the attached symbol. We predict the symbol which gives us the highest probability.

E. Results

We can infer from Table VIII that all our models consistently perform better than the majority baseline. In addition, we observe that the Markov Chain Model performs the best on the CHF-COST dataset and the 4-gram approach performs best on all other datasets. We compare our Markov Chain and MultinomialHMM models only to 4-gram because after a while the results

⁶Higher order Markov models become increasingly difficult to deal with, due to increasing complexity.

for n-gram stabilize and 4-gram is a good representation of what the results for the n-gram technique would look like (Table V).

IV. TECHNIQUES AND RESULTS - INCLUDING FEATURES FROM DISCHARGE RECORDS

In the earlier sections, we obtained results on the OSHPD-SPICE data. This data did not contain any information from the patient discharge records other than symbols for the problem we were trying to predict. In this section, we incorporate features like age, gender, length of stay and severity code into our data, and run our models (Markov Chain Model and Multinomial HMM) on this modified data.

We do this in two fundamentally different ways, depending on the intrinsic nature of the features. Some features, such as age and gender, are stable. Changes in the values of these features across discharge records for the same patient are rare. To incorporate these features in our approach, we build a set of Markov Chain models and Hidden Markov models, where each model is trained and invoked for a specific age and gender category. We provide more details in section IV-A.

Other features are very dynamic, in the sense that their value changes easily from admission to admission, even for the same patient. Examples include, the length of stay (LOS) and the severity of the patient's medical condition. We explain the integration of such features into our Markov Chain Models in section IV-B.

A. Static Features

1) *Age*: We use the static feature age of the patients, and separate the original OSHPD data into two sets: patients whose age is greater than or equal to 65 years, and patients whose age is less than 65 years. We choose the number 65 because 65 years of age is a medically meaningful threshold to separate people of similar health conditions ⁷. We then convert these two sets into OSHPD-SPICE format (as explained in section II-C), and train Markov Chain Models (as explained in section III-D) and Multinomial HMMs (as explained in section III-C) on these datasets.

2) *Gender*: We use the static feature gender in combination with the age feature and separate the OSHPD data further into 4 sets: females ≥ 65 , males ≥ 65 , females < 65 , males < 65 . We convert these sets into OSHPD-SPICE format as before and run Markov Chain model and Multinomial HMM on them. As explained earlier, we calculate the weighted average of results on all four datasets to compute the final accuracy of a model.

⁷<https://www.cbo.gov/publication/44661>

Thus, to summarize we train two models (HMM and Markov Chain) for each category of age: patients ≥ 65 , and patients < 65 each (4 models total). Similarly, we train these two models (HMM and Markov Chain) for each category of gender, i.e females ≥ 65 , males ≥ 65 , females < 65 , males < 65 (8 models total). We do this for all 6 prediction tasks.

To obtain the results for age and gender for each model (HMM and Markov Chain), we take the weighted average of the results for all the categories (2 for age and 4 for gender) to compute the final accuracy.

Eg: If the accuracy of the model trained for age ≥ 65 is x_1 , and the number of sequences in the test file is y_1 , and if the accuracy of the model trained for age < 65 is x_2 , and the number of sequences in the test file is y_2 , then the final accuracy of the model for age is:

$$(x_1*y_1 + x_2*y_2) / (y_1+y_2)$$

Table IX gives the number of sequences (equal to number of patients) for each static feature.

Thus, we finally have four models: Markov Chain-age, HMM-age, Markov Chain-age-gender, HMM-age-gender.

B. Dynamic Features

For this experiment, we use the original OSHPD data which contained all the features instead of the OSHPD-SPICE data. We included the length of stay and severity code features in our previous Markov Chain Model for all six prediction problems. We did not use these features in the Multinomial HMMs since we used a library implementation of HMM and it was not possible to include additional features to it. The field severity code represents whether the patient has a complication or a major complication when admitted to the hospital. This field has three symbols (0,1,2) ⁸.

In our new version of the Markov Chain Model, we form the state transition matrix where:

Number of rows and columns = Number of symbols in LOS * Number of symbols in severity code = $6 * 3 = 18$

There will be N such matrices corresponding to each of the symbols (N) in the prediction task we are trying to solve.

We now consider an example to show how this new Markov chain model could be applied to solve the 30day risk of readmission problem.

We know that our problem is a classification task with possible values (0,1). Thus we will construct two state

⁸ 0 : Patient does not have a complication/major complication, 1 : Patient has a major complication (life threatening), 2 : Patient has a complication

TABLE VIII
RESULTS-WITHOUT FEATURES (ACCURACY)

Model	ALL-30	CHF-30	ALL-LOS	CHF-LOS	ALL-COST	CHF-COST
Majority Baseline	0.648	0.623	0.255	0.258	0.356	0.325
4-gram	0.659	0.640	0.273	0.269	0.418	0.348
Markov Chain	0.648	0.623	0.266	0.268	0.398	0.352
MultinomialHMM	0.652	0.634	0.222	0.238	0.405	0.341

TABLE IX
NUMBER OF SEQUENCES (PATIENTS) FOR EACH FEATURE CATEGORY

Feature	Cohort	Number of Sequences
Age $\geq$ 65	ALLCAUSE CHF	20518 12474
Age < 65	ALLCAUSE CHF	22957 6060
Age $\geq$ 65, Female	ALLCAUSE CHF	9134 5579
Age $\geq$ 65, Male	ALLCAUSE CHF	11384 6895
Age < 65, Female	ALLCAUSE CHF	11432 3314
Age < 65, Male	ALLCAUSE CHF	11525 2746

transition matrices: one for class label 1, where we will store the transition frequencies of training instances of all the patients who have class label 1 in the last record, and another matrix for class label 0, where we will store the transition frequencies of training instances of all the patients who have class label 0 in the last record. We map out all 18 combinations of LOS and severity codes to the transition matrix indices, i.e we store all 18 possibilities in a dictionary along with the corresponding index it maps to. We use the target variable from the data to decide which matrix we want to store our frequencies in. We use the values of LOS and severity code to decide the transition where we want to increment the frequency.

When we want to predict a new instance, we obtain the last transition (LOS-Severity code) from the patient data, and look at that row in the matrix which corresponds to a possible symbol in the problem we are trying to predict (we look at all symbols in turn). Now, we obtain the maximum probability from that row, and store it corresponding to the symbol it was obtained for. We obtain maximum probabilities in this way for all symbols, and we predict the symbol that gives us the maximum probability out of all the symbols.

C. Final Results

Table X shows the final results of all the models tested in this project. We observe that 4-gram performs best for both the 30day problems (ALL-30 and CHF-30). The Markov Chain models with age (and gender) feature

perform the best for the length of stay (LOS) problems. For the ALL-COST problem 4-gram performs the best and for CHF-COST the Markov Chain model (without any features) works the best.

Table XI shows the precision and recall values for the top three models: 4-gram, Markov Chain Model without features and Markov Chain Model with Age feature. We use the sklearn metrics `precision_score` and `recall_score` libraries ⁹ (with `attribute average='weighted'`) to compute the metrics for each label, and find their average, weighted by the number of true instances for each label.

V. CONCLUSION

We observe from Table X that 4-gram performs best for both the 30day problems (ALL-30 and CHF-30). The Markov Chain models with age (and gender) feature perform the best for the length of stay (LOS) problems. For the ALL-COST problem 4-gram performs the best and for CHF-COST the Markov Chain model (without any features) works the best. Thus, we note that every problem has its own peculiarity which makes different models work better for each problem. Eg: We notice that for problems with the same number of possible symbols, the same models works: 4-gram works for both 30day problems, Markov Chain with age feature works for both LOS problems. For cost problems although two different models work for each, we can say that the

⁹http://scikit-learn.org/stable/modules/generated/sklearn.metrics.precision_score.html

TABLE X

RESULTS - ALL MODELS (ACCURACY). IN THIS PROJECT, WE USE THE OSHPD DATA AND CONVERT IT INTO SEQUENCES OF RESPONSE VARIABLE ONLY (OSHDP-SPICE). WE SPLIT THIS DATASET INTO 80% TRAIN AND 20% TEST DATA AND USE THIS DATASET FOR ALL OUR PROBLEMS, EXCEPT MARKOV CHAIN WITH LOS-SEVERITY CODE FEATURE. WE USE THE COMPLETE OSHPD DATA FOR THIS PROBLEM. WE USE ONLY SEQUENCES OF LENGTH FOUR OR MORE FOR ALL PREDICTION TASKS.

Model	Feature	ALL-30	CHF-30	ALL-LOS	CHF-LOS	ALL-COST	CHF-COST
Majority Baseline	-	0.648	0.623	0.255	0.258	0.356	0.325
4-gram	-	0.659	0.640	0.273	0.269	0.418	0.348
MultinomialHMM	-	0.652	0.634	0.222	0.238	0.405	0.341
MultinomialHMM	Age	0.649	0.625	0.254	0.275	0.373	0.327
MultinomialHMM	Age-Gender	0.643	0.621	0.262	0.275	0.384	0.316
Markov Chain	-	0.648	0.623	0.266	0.268	0.398	0.352
Markov Chain	Age	0.632	0.617	0.282	0.289	0.380	0.336
Markov Chain	Age-Gender	0.643	0.617	0.282	0.289	0.380	0.337
Markov Chain	LOS-Severity Code	0.648	0.623	0.278	0.288	0.315	0.292

TABLE XI

PRECISION AND RECALL VALUES FOR TOP THREE MODELS.

Model	Metric	Feature	ALL-30	CHF-30	ALL-LOS	CHF-LOS	ALL-COST	CHF-COST
4-gram	Precision	-	0.539	0.555	0.240	0.236	0.395	0.345
4-gram	Recall	-	0.230	0.223	0.273	0.269	0.418	0.348
Markov Chain	Precision	-	0.0	0.0	0.137	0.115	0.285	0.257
Markov Chain	Recall	-	0.0	0.0	0.266	0.268	0.398	0.352
Markov Chain	Precision	Age	0.230	0.238	0.110	0.096	0.300	0.294
Markov Chain	Recall	Age	0.239	0.246	0.283	0.283	0.38	0.335

results for 4-gram are pretty close to the best model. We can conclude that the reason different models work for different problems may be due to the different number of possible symbols in each problem. Thus, we cannot, in this study, bring forward one model that will perform the best on all our problems.

VI. FUTURE WORK

An interesting direction for future research on this project would be to explore how we could incorporate dynamic features from the discharge records into the 4-gram method (since it works the best for most problems). Additionally, since we used a library implementation of Multinomial HMM, we could not incorporate dynamic features. It would be interesting to see how adding dynamic features to a fundamental implementation of HMM would perform. Further, we could experiment with different features added to the Markov Chain Model.

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VII. APPENDIX A - MODELS TRAINED

This section presents trained models of the Markov Chain and Multinomial Hidden Markov Models for each prediction task.

A. Markov Chain - Without Features

1) *ALL-30*:

$$\begin{bmatrix} \begin{bmatrix} 0.30824335 & 0.09684529 \\ 0.12562398 & 0.07917715 \end{bmatrix} \\ \begin{bmatrix} 0.1160521 & 0.10011561 \\ 0.05999404 & 0.11395017 \end{bmatrix} \end{bmatrix}$$

2) *CHF-30*:

$$\begin{bmatrix} \begin{bmatrix} 0.28802487 & 0.09572854 \\ 0.12051988 & 0.07176178 \end{bmatrix} \\ \begin{bmatrix} 0.13207033 & 0.10920881 \\ 0.06672078 & 0.11596867 \end{bmatrix} \end{bmatrix}$$

3) *ALL-LOS*:

0.01202023	0.0046786	0.00301976	0.00361686	0.00169989	0.000683
0.00901062	0.00485464	0.00353773	0.00436672	0.00201219	0.00078075
0.00654437	0.00350387	0.00318014	0.00431594	0.00214633	0.000851
0.00889848	0.00439	0.00432356	0.00750286	0.00417841	0.00170623
0.00480344	0.00206678	0.00206635	0.00412974	0.00325928	0.00157124
0.00219119	0.00081334	0.00085185	0.00170835	0.00153993	0.00132665
0.00534637	0.00893487	0.00331006	0.00401972	0.00179975	0.00072278
0.00470187	0.01114087	0.0044412	0.00540815	0.0024032	0.0008747
0.00327747	0.00885532	0.00422369	0.00555753	0.0025238	0.00092675
0.00397317	0.01176125	0.00556557	0.00947696	0.00494901	0.0018425
0.00182684	0.00580551	0.0025822	0.00496128	0.00376328	0.00164233
0.00073082	0.00241547	0.00098218	0.00191359	0.00162245	0.00133807
0.0037764	0.00352291	0.00603614	0.00360797	0.00167746	0.00069739
0.00347975	0.00469891	0.00846811	0.00530786	0.00241082	0.00089332
0.00268842	0.00400618	0.00916296	0.00616775	0.00277898	0.00099953
0.00353646	0.00525665	0.01315222	0.01117388	0.00571791	0.00217976
0.00168465	0.00240278	0.00644365	0.00585164	0.00442131	0.00193898
0.00064365	0.00092633	0.00272058	0.00226609	0.00196352	0.00163514
0.00500868	0.00480047	0.00402057	0.01052897	0.0031264	0.00134738
0.0046968	0.00643138	0.00590496	0.01500571	0.00455292	0.00175828
0.00382887	0.00579451	0.00661292	0.01756422	0.00546062	0.00214845
0.00568575	0.00865135	0.01058525	0.03592654	0.01250307	0.00469426
0.00303965	0.00436926	0.00527104	0.02052939	0.01042529	0.00443824
0.00124371	0.00170666	0.00214845	0.00856629	0.0045051	0.0037819
0.0029876	0.00274131	0.00241293	0.00387415	0.00493801	0.00113495
0.00272566	0.00376709	0.00348271	0.0056794	0.00687614	0.00150861
0.00232195	0.00336084	0.00392535	0.00687064	0.00840675	0.00185985
0.00358301	0.00537557	0.00667048	0.01508992	0.01951166	0.00447886
0.00217172	0.00320596	0.00406077	0.01035843	0.01802167	0.00490077
0.00099742	0.00136516	0.0017202	0.0043316	0.00894037	0.00435022
0.00187677	0.00167069	0.00148576	0.00241759	0.00167661	0.00253946
0.00162837	0.0022534	0.00212687	0.00345732	0.00234395	0.00344759
0.00139774	0.00202107	0.00259617	0.00433583	0.00291862	0.00437476
0.00227836	0.00327324	0.00423977	0.00985697	0.00728027	0.00994795
0.00144556	0.00213618	0.00275655	0.00696627	0.00767424	0.01050358
0.00092802	0.00125852	0.0015687	0.00411917	0.00495282	0.01158351

4) CHF-LOS:

0.0084913	0.00379715	0.00267726	0.00333305	0.00164727	0.00068881
0.00705591	0.00416219	0.00320556	0.00431719	0.00212604	0.0008484
0.00562877	0.00316337	0.00298544	0.00438506	0.0022361	0.0009392
0.00810425	0.00426766	0.00438323	0.00766125	0.00452081	0.0019747
0.00480697	0.00205725	0.00214621	0.00444652	0.00332938	0.00165277
0.00227645	0.00085115	0.00092819	0.00188023	0.00164818	0.00132808
0.00427775	0.00693117	0.00290289	0.00379623	0.0016757	0.00071999
0.00392372	0.00841885	0.00400902	0.00517384	0.00230122	0.00094103
0.00286345	0.00757595	0.0038531	0.00543983	0.00259747	0.0009502
0.0035862	0.01077693	0.0054224	0.00914984	0.00499958	0.00198846
0.00174174	0.0055022	0.00255803	0.00495464	0.0036238	0.00163167
0.00070715	0.00234799	0.00105109	0.0019958	0.00166653	0.00130424
0.00328261	0.0031276	0.00501242	0.00340918	0.00160507	0.00070715
0.00306432	0.00414568	0.00739252	0.00515	0.00240578	0.00095204
0.00239385	0.00368158	0.00775113	0.00591952	0.00275064	0.00100248
0.00328903	0.00509405	0.0121738	0.01074666	0.00560033	0.00229572
0.00158765	0.0023856	0.00618917	0.00575809	0.00415852	0.00188757
0.00063102	0.00092085	0.00269469	0.00233515	0.00196094	0.00161241
0.0045896	0.00472993	0.00400993	0.00966071	0.00319639	0.00141338
0.004563	0.00643406	0.00608644	0.01456399	0.00473726	0.00190958
0.00384484	0.00591768	0.00658356	0.01733205	0.00562877	0.00237459
0.00558933	0.00878297	0.01074758	0.03423578	0.01272687	0.00511331
0.00301571	0.00452631	0.00543708	0.02053487	0.01000191	0.00445202
0.00127947	0.00181878	0.00231589	0.008993	0.00453823	0.00379807
0.00309642	0.00299278	0.00273596	0.00417961	0.00487209	0.00123453
0.00296985	0.00416586	0.00396499	0.00640379	0.00718982	0.00167295
0.00255069	0.003832	0.00433278	0.00781717	0.00894531	0.00210953
0.00390354	0.006036	0.00754477	0.01615531	0.02021477	0.0049473
0.00219391	0.00345229	0.00437222	0.01078518	0.01731279	0.00481339
0.00104743	0.00152069	0.00193985	0.00469874	0.00899117	0.00421905
0.00203248	0.00197103	0.00183254	0.00285245	0.00191141	0.0025672
0.00192426	0.00262957	0.00257087	0.00434654	0.00281576	0.00375404
0.00166011	0.00247181	0.00306248	0.00541415	0.00358436	0.00497757
0.00269377	0.00404479	0.00525364	0.01179225	0.00869308	0.01124378
0.00164818	0.00256445	0.00331104	0.00818588	0.00827485	0.01129606
0.00098964	0.00136385	0.00183712	0.00469874	0.00516742	0.01164459

5) *ALL-COST*:

$$\begin{bmatrix}
\begin{bmatrix} 0.05413359 & 0.01414933 & 0.00682668 & 0.00355976 \\ 0.02112836 & 0.0117423 & 0.00713095 & 0.00413486 \\ 0.01175246 & 0.00727356 & 0.00672258 & 0.00453688 \\ 0.0072524 & 0.00420299 & 0.00465113 & 0.00516064 \end{bmatrix} \\
\begin{bmatrix} 0.01938445 & 0.02082155 & 0.00780676 & 0.00464732 \\ 0.01266483 & 0.02910439 & 0.01346463 & 0.00811822 \\ 0.00739501 & 0.02317188 & 0.01458859 & 0.01035725 \\ 0.00428212 & 0.0159965 & 0.01056376 & 0.01218198 \end{bmatrix} \\
\begin{bmatrix} 0.01247144 & 0.01050451 & 0.01363983 & 0.00564137 \\ 0.00995607 & 0.01737563 & 0.0271874 & 0.01166275 \\ 0.00715888 & 0.01597195 & 0.03676769 & 0.01819786 \\ 0.00481152 & 0.01108807 & 0.03174036 & 0.02377236 \end{bmatrix} \\
\begin{bmatrix} 0.01108934 & 0.0099548 & 0.00852701 & 0.01299617 \\ 0.00936786 & 0.01742768 & 0.01862061 & 0.02819837 \\ 0.00762353 & 0.01783858 & 0.02858219 & 0.04830983 \\ 0.00651057 & 0.01559913 & 0.02999052 & 0.08252873 \end{bmatrix}
\end{bmatrix}$$

6) *CHF-COST*:

$$\begin{bmatrix}
\begin{bmatrix} 0.04936199 & 0.01722231 & 0.00984892 & 0.00570316 \\ 0.02512217 & 0.01352507 & 0.00932153 & 0.00547753 \\ 0.01610791 & 0.00924815 & 0.00837956 & 0.00556558 \\ 0.01008556 & 0.00558484 & 0.00568849 & 0.00542892 \end{bmatrix} \\
\begin{bmatrix} 0.02502311 & 0.02524049 & 0.01067348 & 0.0064727 \\ 0.01569333 & 0.0285947 & 0.01339299 & 0.00812091 \\ 0.00995256 & 0.02280715 & 0.01359569 & 0.00935179 \\ 0.00587652 & 0.01506688 & 0.00953248 & 0.00934629 \end{bmatrix} \\
\begin{bmatrix} 0.01941075 & 0.01411024 & 0.01805421 & 0.00726699 \\ 0.01340858 & 0.01752498 & 0.02585135 & 0.01052306 \\ 0.00960953 & 0.01520538 & 0.02992923 & 0.01373144 \\ 0.00632778 & 0.00995898 & 0.02398759 & 0.01528334 \end{bmatrix} \\
\begin{bmatrix} 0.01974553 & 0.0150999 & 0.01208231 & 0.01679764 \\ 0.01410015 & 0.0186935 & 0.01842751 & 0.0256037 \\ 0.01094864 & 0.01744244 & 0.02293189 & 0.03596534 \\ 0.00784116 & 0.01348838 & 0.021114 & 0.04916571 \end{bmatrix}
\end{bmatrix}$$

*B. Multinomial HMM - Without Features*1) *ALL-30*:

Transition Matrix:

$$\begin{bmatrix} 0.79426227 & 0.20573773 \\ 0.1953389 & 0.8046611 \end{bmatrix}$$

Emission Matrix:

$$\begin{bmatrix} 0.86162129 & 0.13837871 \\ 0.31028938 & 0.68971062 \end{bmatrix}$$

2) *CHF-30*:

Transition Matrix:

$$\begin{bmatrix} 0.92142695 & 0.07857305 \\ 0.08734935 & 0.91265065 \end{bmatrix}$$

Emission Matrix:

$$\begin{bmatrix} 0.74997352 & 0.25002648 \\ 0.25090457 & 0.74909543 \end{bmatrix}$$

3) *ALL-LOS*:

Transition Matrix:

$$\begin{bmatrix} 0.57168681 & 0.42831319 \\ 0.27202431 & 0.72797569 \end{bmatrix}$$

Emission Matrix:

$$\begin{bmatrix} 0.31855639 & 0.23035144 & 0.14139892 & 0.0167657 & 0.09015671 & 0.20277084 \\ 0.00627153 & 0.10738942 & 0.16555566 & 0.44269064 & 0.24342777 & 0.03466498 \end{bmatrix}$$

4) *CHF-LOS*:

Transition Matrix:

$$\begin{bmatrix} 0.57669198 & 0.42330802 \\ 0.4529103 & 0.5470897 \end{bmatrix}$$

Emission Matrix:

$$\begin{bmatrix} 0.08261566 & 0.23432495 & 0.16268252 & 0.25525974 & 0.12652719 & 0.13858995 \\ 0.15747716 & 0.05796858 & 0.14834772 & 0.31081232 & 0.25494244 & 0.07045178 \end{bmatrix}$$

5) *ALL-COST*:

Transition Matrix:

$$\begin{bmatrix} 0.98459635 & 0.01540365 \\ 0.0597789 & 0.9402211 \end{bmatrix}$$

Emission Matrix:

$$\begin{bmatrix} 0.05724335 & 0.20172116 & 0.33105221 & 0.40998328 \\ 0.48353847 & 0.31759401 & 0.14819466 & 0.05067286 \end{bmatrix}$$

6) *CHF-COST*:

Transition Matrix:

$$\begin{bmatrix} 0.91093437 & 0.08906563 \\ 0.03975916 & 0.96024084 \end{bmatrix}$$

Emission Matrix:

$$\begin{bmatrix} 0.48389484 & 0.30124029 & 0.15369595 & 0.06116892 \\ 0.08056214 & 0.22426395 & 0.33149914 & 0.36367477 \end{bmatrix}$$

VIII. APPENDIX B - ACM-BCB POSTER AND PAPER

Sequence Based Prediction of Hospital Readmissions

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ABSTRACT

We propose an N-gram based approach to make predictions about the next event in a sequence of hospital admissions. To predict the value of a variable in the next admission (e.g. *length of stay*), we use only the sequence of values of the same variable for all hospital admissions of the same patient so far (e.g. *the length of all previous hospital stays*) and no other information from the discharge record. We validate our method on inpatient data from the California Office of Statewide Health Planning and Development (OSHPD), for the prediction of 3 kinds of variables: whether the patient will be readmitted within 30 days, and what the associated length of stay and cost of the next admission will be. Our results show that in all cases 4-gram, 5-gram and 6-gram methods make more accurate predictions than the majority baseline method, by a margin that depends on the problem at hand as well as on the cohort, i.e. the patient population as a whole (“ALL-Cause”) versus a congestive heart failure (CHF) cohort.

Categories and Subject Descriptors

D.2.8 [Life and Medical Sciences]: Health; I.5.2 [Pattern Recognition]: Design Methodology—*Classifier design and evaluation*

1. INTRODUCTION AND BACKGROUND

Sequences are an important type of data which occur frequently in several scientific, medical, security, and other domains. Sequence data mining provides helpful tools for exploring useful knowledge hidden in large sequence datasets. Many prediction problems in clinical care settings are inherently sequence prediction problems: given a sequence of previous patient encounters (e.g. hospital visits) the task is to predict one or more aspects about future encounters with the same patient.

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The problem of hospital readmissions is very severe in the U.S. Currently 1 in 5 patients is readmitted to the hospital within 30 days of being discharged. Measures that can reduce the number of unnecessary, lengthy and costly hospital readmissions are therefore very valuable [2]. The ability to predict them accurately provides many benefits for accountable care, now a global issue and foundation for the U.S. government mandate under the Affordable Care Act. While predicting 30-day readmission has been identified as one of the key problems for the healthcare domain, not many solutions are known to be effective [4]. Furthermore, uncertainty in length of hospital stay is a major deterrent to effective scheduling for admission of elective patients. A model to predict the Length of Stay (LOS) for hospitalized patients can be an effective tool for healthcare providers, as it will enable early interventions to prevent complications, among other things [1]. However, the ability to risk stratify for LOS based on patient admission and hospital characteristics is limited, and particularly challenging for congestive heart failure (CHF) patients. Readmissions and prolonged hospital stay act as substantial contributors to rising healthcare costs [3]. Alongside of predicting *30-day readmission* and *LOS*, in this contribution we also investigate algorithm performance for forecasting the *cost* of hospital admissions. Previously proposed cost prediction models were primarily focused on ‘general’ healthcare costs as apposed to hospital admissions, and were often rule based and regression models. Additionally, the development of healthcare cost prediction models using data mining techniques has been more recent (e.g. [7]). It is a common practice to make predictions about the *next* hospital admission using the discharge record of the *current* admission. Leveraging the temporal dimension of the entire sequence of prior hospital admissions has remained largely unexplored so far.

In this contribution we explore the use of a statistical N-gram based approach to predict variables about the next event in a sequence of admissions of the same patient. These models exploit probability distributions over sequences of events and have been successfully utilized in several domains (e.g., information retrieval [6]). We use an N-gram approach to predict whether the patient will be readmitted within 30 days, and what the associated length of stay and cost of the next hospital admission will be. The only input features we use are the values of these same three variables (30-Day, LOS and cost) for all prior admissions of the same patient. We validate our method on inpatient data from the California Office of Statewide Health Planning and Development (OSHPD), on the patient population as a whole (ALL-Cause), as well as on a congestive heart failure (CHF) cohort. To the best of our knowledge, our study is the first to explore the potential of next event prediction in symbolic sequences extracted from a large inpatient dataset.

Table 1: Statistics of the train and test sequence datasets for both cohorts (“ALL-Cause” and “CHF”) for the 3 prediction problems under study, i.e. whether the next admission will be within 30 days or not, and what the length and cost of the next admission will be.

Problem	# Seq	Sequence Length			Symbols
		Min	Max	Avg (SD)	
ALL-30 train	2,538,134	1	333	2.5 (3.2)	0.1
ALL-30 test	281,255	1	174	2.5 (3.2)	0.1
CHF-30 train	530,594	1	333	3.9 (4.6)	0.1
CHF-30 test	58,814	1	174	3.9 (4.6)	0.1
ALL-LOS train	2,538,134	1	333	2.5 (3.2)	1,2,3,4,5,6
ALL-LOS test	281,255	1	174	2.5 (3.2)	1,2,3,4,5,6
CHF-LOS train	448,067	1	255	3.9 (4.4)	1,2,3,4,5,6
CHF-LOS test	58,814	1	174	3.9 (4.6)	1,2,3,4,5,6
ALL-COST train	2,125,302	1	255	2.5 (3.2)	1,2,3,4
ALL-COST test	235,879	1	155	2.5 (3.1)	1,2,3,4
CHF-COST train	448,067	1	255	3.9 (4.4)	1,2,3,4
CHF-COST test	49,630	1	164	3.9 (4.5)	1,2,3,4

2. METHODS AND RESULTS

We requested non-public OSHPD data for the years 2009-2013. The dataset is a collection of records in tabular format with each row corresponding to one hospital discharge record of one patient. After applying a series of data preprocessing steps (see [5] for a description of similar steps) we extracted sequences for our prediction problems as shown in Table 1. For instance, for the 30-day readmission problem in the general population (ALL-30), our pre-processed dataset contains 2,819,389 sequences. Each sequence corresponds to a unique patient. The symbols in the sequence are 1s and 0s, denoting respectively that the next admission of the patient was within 30 days of the previous admission, or not. For instance, the sequence “1,0,0,1,0,1,1” means that the patient had 7 hospital readmissions, 4 of which were within 30 days. Out of the 2,819,389 sequences we set aside approx. 10% for testing (281,255 sequences) and used the rest (2,538,134 sequences) for training, i.e. to construct the N-gram dictionary (see below).

The other sequence datasets in Table 1 were constructed in a similar way. The sequences in the CHF datasets are limited to those of patients who have CHF as a primary or secondary diagnosis in at least one of their records¹. For LOS we created 6 symbols, corresponding to the discretization used in the well known LACE index [8], i.e. 1, 2, 3, [4-6], [7-13], and 14+ days. For cost we created 4 symbols, each corresponding to a quartile in the data². We add the symbol -1 in front of all the training and test sequences to mark the beginning of a sequence. To predict the next value for a given test sequence using N-grams, probabilities are estimated during the training phase, which are then used during the test phase to predict the next most likely symbol.

Training Phase: The frequencies of all possible grams (1-gram, 2-gram, etc.) in the training sequences are computed and stored in a dictionary. E.g. the training sequence “-1, 2, 1, 0, 3, 2, 0” will lead to the following frequency increments: (-1):1, (2):2, (1):1, (0):2, (3):1, (-1,2):1, (2,1):1, (1,0):1, (0,3):1, (3,2):1, (2,0):1, (-1,2,1): 1, (2,1,0):1, (1,0,3):1, (0,3,2):1, (3,2,0):1, etc.

Testing Phase: Let L denote the length of the test sequence. In the N-gram approach, we take the last M symbols of the test sequence, with $M = \min(L, N-1)$, and we complete this to the most probable $M+1$ gram. I.e., given the last M symbols $(a_1, \dots, a_M)$, we look up the most frequent $M+1$ gram $(a_1, \dots, a_M, b)$ and predict b as the next symbol in the sequence. E.g., the 5-gram approach will consult all 5-grams from the dictionary whose prefix matches the last 4 symbols of the test sequence. It will only resort to grams of a smaller length if the test sequence has less than 4 symbols, and it will never consult any grams of length > 5 .

The results for all prediction tasks are presented in terms of accuracy in Table 2. We tested the accuracy of N-gram models from $N = 1$ to 6, and contrasted their performance with random guessing as well with a majority baseline technique that systematically predicts the most frequently occurring symbol in the

¹Using the ICD-9-CM codes for CHF: 398.91 and 428.XX.

²Cost ranges (in dollars) for ALL-Cause: 0-19,080, 19,080-35,690, 35,690-70,450, 70,450+, and for CHF: 0-27,660, 27,660-49,810, 49,810-98,500, 98,500+.

Table 2: Performance comparison for 30-Day, LOS and cost of next hospital admission prediction task.

Method	ALL 30	CHF 30	ALL LOS	CHF LOS	ALL COST	CHF COST
Random	50.00	50.00	16.67	16.67	25.00	25.00
Majority	76.93	69.07	22.22	25.47	25.79	26.08
2-gram	76.93	69.07	25.63	25.47	33.26	29.87
3-gram	77.01	69.52	26.20	25.93	33.79	30.05
4-gram	77.02	69.66	26.22	26.03	33.92	30.18
5-gram	77.10	69.74	26.25	26.03	33.95	30.12
6-gram	77.17	69.74	26.19	25.78	33.97	30.16

training data. For instance, for 30-day readmission, the majority method always predicts 0 as the next symbol, since the majority of readmissions is not within 30 days. Three key observations can be made: (1) the N-gram models consistently perform better than the majority baseline technique for all prediction tasks, but marginally better for CHF-30-Day and CHF-LOS prediction; (2) accurately predicting variables related to the next hospital event using an N-gram approach appears easier for the patient population as a whole (ALL) than the specific cohort (CHF); (3) increasing longitudinal information (longer sequence, hence bigger N) not necessarily seems to further improve the performance. The performance of the 4-gram models is comparable to, and sometimes even better than, that of the 5- and 6-gram models.

3. CONCLUSION

Being able to stratify patients according to 30-day readmission risk, anticipated length and cost of stay can guide clinicians in discharge planning and intervention recommendation, leading to an increase of quality of care, and a decrease of healthcare cost. In this contribution we explored the use of a statistical N-gram based approach to predict variables about the next event in a sequence of admissions of the same patient. Our preliminary results show that simple N-gram methods make more accurate predictions than the majority baseline method, indicating that there is value in taking sequential information into account. In future work we plan to enrich the simple symbolic sequences considered in this study with features from hospital discharge records.

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Sequence Based Prediction of Hospital Readmissions



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Problem

Importance of the Problem:



There are about 34 million hospital admissions annually in the U.S., with 1 in 5 patients being readmitted to the hospital within 30 days of being discharged.

Being able to stratify patients according to 30-day hospital readmission risk, anticipated length and cost of stay can guide clinicians in discharge planning and intervention recommendation, leading to an increase of quality of care, and a decrease of healthcare cost.

Research Goals:

Given the admission history of a patient, predict:

Task to predict	Abbrev.
Will the patient get readmitted within 30 days? Yes/No	30
How long will the patient stay in the hospital if readmitted? (Days)	LOS
What will be the cost of this readmission? (\$)	COST

Cohorts:

Cohort Description	Abbrev.
Patient population considered as a whole	ALL
Congestive Heart Failure patients only	CHF

Example:

- Patient Anna has the following admission history. Given the history, we need to predict whether she will be readmitted within 30 days, her length of stay and cost.

Admission Date	30Day	LOS	COST
Jan 22, 2009	0	6	\$75099
Mar 30, 2012	0	2	\$37531
Apr 15, 2012	1	5	\$40657
Oct 1, 2013	0	1	\$14107
Oct 20, 2013	1	14	\$259616
???	???	???	???

Emphasis on Sequence Based Prediction:

Given a sequence of previous patient encounters (e.g. hospital visits), predict one or more aspects about future encounters for the same patient.

Methods

Data:

- OSHPD data from California-licensed hospitals where each row corresponds to one admission record of a patient (2009-2013).
- After preprocessing, we converted the original OSHPD data into sequences. The only input features used are the values of 30-Day, LOS and COST for all prior admissions of the same patient.
- E.g.: For patient Anna (as mentioned earlier), while predicting risk of readmission(30-Day), we convert her admission records to a sequence of the form: "0, 0, 1, 0, 1". This symbolizes that the patient has 5 hospital admissions, 2 of which were within 30 days. Similarly, we create sequences for LOS, namely "4, 2, 4, 1, 6", and for cost, namely "4, 3, 3, 1, 4".
- The OSHPD data has actual LOS and COST values as shown in Anna's admission history. We convert these values to symbols as shown in the table below.

LOS Symbols (Days)	LOS Values	COST Symbols	COST Values	COST Values (CHF \$SS)
1	1	1	0-19,080	0-27,660
2	2	2	19,080-35,690	27,660-49,810
3	3	3	35,690-70,450	49,810-98,500
4	4-6	4	70,450+	98,500+
5	7-13			
6	14+			

Data Statistics:

Problem	#Seq	Min-Max Seq Length	Avg Seq Length (SD)
ALL Train	390,900	4-255	7.0 (5.3)
ALL Test	43,475	4-155	7.0 (5.1)
CHF Train	168,576	4-255	7.5 (5.6)
CHF Test	18,534	4-164	7.5 (5.5)

N-gram method:

- We add the symbol -1 in front of all the training and test sequences to mark the beginning of a sequence.
- **TRAINING:** The frequencies of all possible n-grams (1-gram, 2-gram, etc.) in the training sequences are computed and stored in a dictionary. E.g. the training sequence "-1, 2, 1, 0, 3, 2, 0" will lead to the following frequency increments (3-gram approach): (-1,1):1, (1,2):2, (2,1):0, (2,3):1, (-1,2,1):1, (1,0,1):0, (0,2,1):1, (2,0,1):1, (-1,2,0,1):1, (0,2,0,1):1, (0,3,2,1):1, (3,2,0,1):1.
- **TESTING:** Given the last M symbols ($a_1, a_2, \dots, a_M$), we look up the most frequent M+1 gram (3-gram approach in this case) ($a_1, a_2, \dots, a_M, b$) and predict b as the next symbol in the sequence.

Hidden Markov Model (HMM):

- HMM is a statistical Markov Model in which the system being modeled is assumed to be a Markov Process with unobserved (hidden) states.
- In simpler Markov Models (Markov Chain), the state is directly visible to the observer, and therefore the state transition probabilities are the only parameters. In a Hidden Markov Model, the state is not directly visible, but the output, dependent on the state, is visible.
- In this study we obtained results using Multinomial HMMs, i.e. HMMs with discrete emissions. (<http://bmmlearn.readthedocs.io/en/latest/api.html#multinomialhmm>)



Results

Observations:

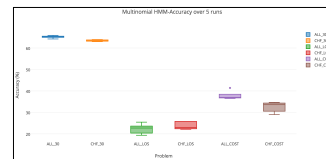
- The N-gram models consistently perform best for all prediction tasks.
- Increasing longitudinal information (longer sequence, hence larger N) will not necessarily improve the performance.
- The performance of the 4-gram models is comparable to, and sometimes even better than, that of 5 and 6-gram models.
- HMM on the other hand, has very fluctuating results. It performs better than the baseline methods for the 30 day risk prediction problem (both cohorts), and the cost prediction ("all cause" cohort).

Results:

- Accuracy results for predicting the last symbol in a sequence of length 4 or more

Method	ALL 30	CHF 30	ALL LOS	CHF LOS	ALL COST	CHF COST
Random	50.00	50.00	16.67	16.67	25.00	25.00
Majority	0.648	0.623	0.255	0.258	0.356	0.325
2-gram	0.621	0.606	0.266	0.258	0.381	0.336
3-gram	0.659	0.640	0.272	0.267	0.408	0.342
4-gram	0.659	0.640	0.273	0.269	0.418	0.348
5-gram	0.662	0.642	0.274	0.269	0.416	0.353
6-gram	0.662	0.642	0.271	0.262	0.417	0.347
HMM	0.652	0.634	0.222	0.238	0.378	0.325

- Multinomial HMM accuracy results across 5 runs



Future Work:

- Comparison with recurrent neural networks (RNNs)
- Incorporating other static and dynamic features from discharge records